

SHUPING LI

Functional Genomics | Immunology | Single-Cell & Computational Genomics

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RESEARCH PROFILE

Ph.D. candidate in Physiology focused on functional genomics and immune-mediated disease, with integrated experimental and computational expertise spanning pooled CRISPR perturbation, HCR-FlowFISH, CRISPR base editing, ATAC-seq, single-cell and bulk transcriptomics, and molecular/functional validation. Research centers on translating disease-associated genetic variation into experimentally testable regulatory mechanisms.

EDUCATION

Ph.D. Candidate, Physiology

08/2021-Present

University of Oklahoma Health Campus, Oklahoma City, OK, USA

Master's Degree, Internal Medicine

08/2018-07/2021

Sun Yat-sen University, Guangdong, China

Bachelor's Degree, Clinical Medicine

09/2013-06/2018

Gannan Medical University, Jiangxi, China

RESEARCH EXPERIENCE

Ph.D. Candidate & Graduate Research Assistant

06/2024-Present

University of Oklahoma Health Campus & Oklahoma Medical Research Foundation, Oklahoma City, OK, USA

Mentor: Patrick M. Gaffney, M.D. | Dissertation: Causal Variant Discovery and Characterization in the SLE-Associated *STAT4* Locus

- Develop and execute a functional genomics strategy to identify candidate cis-regulatory elements (cCREs) within the SLE-associated *STAT4* locus and characterize how disease-associated variants modulate *STAT4* expression.
- Design and conduct pooled CRISPR interference (CRISPRi) screens coupled with HCR-FlowFISH in *STAT4*-high-expressing MOLT-16 and KCL-22s cells to identify regulatory elements controlling *STAT4* expression.
- Generate stable dCas9-KRAB-expressing cell models and establish end-to-end CRISPRi screening workflows encompassing sgRNA library design and construction, lentiviral delivery, library amplification, NGS library preparation, sequencing-based representation QC, workflow optimization, and downstream screen analysis.
- Apply CRISPR base editing coupled with bulk RNA-seq in MOLT-16 and KCL-22s cells to characterize the allelic effects of candidate SLE risk variants on gene expression.
- Functionally characterize CRISPR base-edited models using molecular and cellular assays, including Western blotting, cytokine-stimulated *STAT4* signaling (p-*STAT4*/*STAT4*), cell proliferation assays, Annexin V/PI flow cytometry, and JAK signaling assays.

Graduate Research Assistant

08/2021-04/2024

Department of Physiology, University of Oklahoma Health Sciences Center, Oklahoma City, OK, USA

PI: Xin Zhang, M.D. Ph.D. | Major project: Mechanistic roles of CD82 in breast cancer

- Established a mammary-specific *Cd82* knockout model (*Cd82^{fl/fl}*; *MMTV-Cre*) with the *MMTV-PyMT* transgene and investigated CD82-dependent regulation of breast tumor initiation and growth.
- Used shRNA knockdown, CCK-8 assays, and Incucyte real-time monitoring in T-47D, MDA-MB-468, and MDA-MB-231 cells to characterize CD82-associated suppression of proliferation.
- Established 3D oncosphere assays in murine primary tumor cells and human breast cancer cell lines to investigate cancer stem-cell phenotypes; observed reduced oncosphere formation efficiency and size with CD82/*Cd82*.
- Evaluated glycolytic proteins by Western blotting and identified CD82/*Cd82*-associated reductions in HK2 and, in murine cancer stem cells, LDHA/LDHB.
- Performed transcriptomic analyses of breast cancer cohorts using differential expression, GSEA, WGCNA, GO/KEGG enrichment, and protein-protein interaction analyses to investigate links between CD82 and IL-17-mediated inflammation.

Graduate Researcher

09/2018-07/2021

Department of Nephrology & Edmond H. Fischer Translational Medical Research Laboratory, The Seventh Affiliated Hospital, Sun Yat-sen University, China

Mentors: Yun Chen, Ph.D. and Zhihua Zheng, M.D., Ph.D. | Dissertation: Clinical Investigations in Systemic Lupus Erythematosus

- Characterized Mincle/CLEC4E expression across leukocyte and monocyte subsets from SLE patients and healthy donors using flow cytometry; identified highest expression in intermediate monocytes in SLE.
- Integrated immunophenotyping with clinical data and identified positive correlations between Mincle expression and serum IgG and λ -light-chain levels.
- Performed primary-cell co-culture experiments showing that intermediate monocytes promoted CD4⁺ naïve T-cell differentiation toward Th17 cells; shRNA-mediated Mincle knockdown in THP-1 cells suppressed Th17 induction.
- Investigated lymphoid tyrosine phosphatase transcript variants in peripheral blood using real-time PCR and nucleic-acid silver staining, identifying reduced LYP3/LYP1 levels in SLE patients.
- Led/collaborated on meta-analyses addressing *PTPN22*-C1858T in mycobacterial infection, *FLT3*-TKD in AML prognosis, and vitamin D biomarkers in kidney-transplant outcomes.

Part-Time Research Volunteer

07/2014-07/2018

Physiological Function Laboratory, Gannan Medical University, China

- Established and managed an MCAO rat model and used TTC staining and Western blotting to investigate Celastrol-associated protection from cerebral ischemia-reperfusion injury.

TECHNICAL EXPERTISE

- **Functional genomics:** Pooled CRISPRi screening; sgRNA library design/construction and QC; lentiviral delivery; stable dCas9 cell generation; HCR-FlowFISH; CRISPR base editing; NGS library preparation; sequencing-based library QC and representation analysis.
- **Genomics & single-cell:** ATAC-seq; bulk RNA-seq; scRNA-seq; Cell Ranger; Seurat; Harmony; QC and doublet detection; normalization; clustering; cell-type annotation; batch correction/integration; differential expression; cell-composition analysis.

- **Computational:** R; Python; Linux/HPC; statistical, differential expression, and pathway analysis; WGCNA; multi-cohort integration; public-omics data mining; machine learning (LASSO, Random Forest, SVM-RFE); feature selection; biomarker discovery; predictive modeling.
- **Network & single-cell systems biology:** protein-protein interaction (PPI) network analysis (STRING, Cytoscape); cell-cell communication (CellChat, LIANA, NicheNet); gene-signature scoring (UCell); gene-regulatory network analysis and in silico perturbation (scTenifoldKnn).
- **Molecular & cellular:** 2D/3D mammalian cell culture; cloning; plasmid preparation; PCR/qPCR; Western blot; ELISA; immunofluorescence; Co-IP; PLA; multiparameter flow cytometry/FlowJo; PBMC and immune-cell isolation; stimulation/activation; intracellular cytokine staining; apoptosis/proliferation/viability assays.
- **In vivo & clinical:** Genetically engineered mouse breeding/genotyping; tissue dissociation and primary-cell isolation; MCAO rat model; clinical protocol implementation; recruitment/consent; longitudinal follow-up; biobanking; biospecimen processing; clinical-data analysis.
- **Evidence synthesis:** Systematic review and meta-analysis; study screening; quality assessment; data extraction/curation; heterogeneity/publication-bias assessment; sensitivity analysis; visualization and manuscript preparation.

PEER-REVIEWED PUBLICATIONS

1. Xu J, Lin J, Ma C, Yeerlan J, **Li S**, Zhu W, Zheng Z, Huang M. Serum and urinary golgi membrane protein 1 (GOLM1/GP73/G73) for chronic kidney disease staging. *Front Med (Lausanne)*. 2026;13:1896678.
2. Wang XJ, Richardson MM, Ding Y, Jiang C, Liu S, Zhou B, **Li S**, Chen J, Rubinstein E, Hays FA, Mehla K, Zhang XA. Interaction between CD82 and integrin $\alpha\text{V}\beta 3$ selectively regulates collective movement of tumor cells via endolysosomal trafficking. *Cell Mol Life Sci*. 2026.
3. **Li S**, Li J, Xiao Y, Guo Y, Sui Y, Zhao ZJ, Zheng Z, Chen Y. Upregulated Macrophage-Inducible C-Type Lectin on Intermediate Monocytes Facilitates T Helper 17 Cell Differentiation in Systemic Lupus Erythematosus. *Immunol Lett*. 2026;280:107181.
4. Yu L, Zhang Y, Zheng Y, Zhang D, Chang Z, Zhao Y, Ma L, Xiao Y, **Li S**, Zhao ZJ, Chen C, Guo Y. Homoharringtonine and Gilteritinib Synergistically Induce Apoptosis and Suppress Viability in FLT3-ITD-Positive AML Cells. *Biomedicines*. 2026;14(2).
5. Guan H, Tian C, Lin J, Zhang L, Shi R, Wang W, **Li S**, Sui Y, Lu Y, Cui T, Chen D. Liquid-liquid phase separation-driven molecular subtyping and prognostic modeling in colorectal cancer. *Front Immunol*. 2025;16:1741979.
6. **Li S**, Ouyang H, Cui T, Lin J, Wu K, Zhu E, Sui Y, Huang M. Machine learning reveals sex-biased platelet-associated molecular signatures in systemic lupus erythematosus. *Immunol Lett*. 2026;279:107136.
7. Zhuang X, Ma L, Yu L, Zhao Y, Zhang D, Li C, Liu C, Xiao Y, Chang Z, **Li S**, Chen C, Chen Y, Zhou G, Zhao ZJ, Guo Y. Ultrasensitive Detection of Rare Mutations via Amplifying-Cleaving-Enriching in Acute Myeloid Leukemia. *Biomedicines*. 2025;13(12).
8. Ouyang H, Su J, Lin J, **Li S**, Sui Y, Xie C, Huang M. Inflammatory Bowel Disease and Novel Cell Death: Bibliometric Analysis and Study Trend Projections Based on Burst Detection. *Clin Exp Pharmacol Physiol*. 2025;52(12):e70088.

9. Zhang D, Guo Y, Peng Z, Xiao Y, Chang Z, Yu L, Zhao Y, Zhang Q, Ma L, **Li S**, Li CK, Leung KT, Zhao ZJ, Chen C, Chen Y. FLT3L-based drug conjugate effectively targets chemoresistant leukemia stem cells in acute myeloid leukemia. *Cell Rep Med*. 2025;6(10):102365.
10. Wu K, Zhu E, Chen J, Kuang Q, Lin J, Zhao S, Xu X, **Li S**, Sui Y, Huang M, Zhang Y. Overexpression of GPX4 in diabetic rat kidney alleviates renal injury induced by ferroptosis. *Biometals*. 2025;38(4):1281-97.
11. Lin J, Zhang Y, Guan H, **Li S**, Sui Y, Hong L, Zheng Z, Huang M. Myricitrin inhibited ferritinophagy-mediated ferroptosis in cisplatin-induced human renal tubular epithelial cell injury. *Front Pharmacol*. 2024;15:1372094.
12. Min J, Yang S, Cai Y, Vanderwall DR, Wu Z, **Li S**, Liu S, Liu B, Wang J, Ding Y, Chen J, Jiang C, Wren JD, Csiszar A, Ungvari Z, Greco C, Kanie T, Peng J, Zhang XA. Tetraspanin Tspan8 restrains interferon signaling to stabilize intestinal epithelium by directing endocytosis of interferon receptor. *Cell Mol Life Sci*. 2023;80(6):154.
13. Zeng S, Yang Y, **Li S**, Hoher CF, Chu C, Wang Z, Zheng Z, Krämer BK, Hoher B. 25(OH)D-but not 1,25(OH)(2)D-Is an independent risk factor predicting graft loss in stable kidney transplant recipients. *Front Med (Lausanne)*. 2023;10:1141646.
14. **Li S**, Li N, Chen Y, Zheng Z, Guo Y. FLT3-TKD in the prognosis of patients with acute myeloid leukemia: A meta-analysis. *Front Oncol*. 2023;13:1086846.
15. Ding Y, Chen J, **Li S**, Wren JD, Bajpai AK, Wang J, Tanaka T, Rice HC, Hays FA, Lu L, Zhang XA. EWI2 and its relatives in Tetraspanin-enriched membrane domains regulate malignancy. *Oncogene*. 2023;42(12):861-8.
16. Sui Y, **Li S**, Fu XQ, Zhao ZJ, Xing S. Bioinformatics analyses of combined databases identify shared differentially expressed genes in cancer and autoimmune disease. *J Transl Med*. 2023;21(1):109.
17. **Li S**, Wang X, Zhao Y, Yang J, Cui T, Zhao ZJ, Chen Y, Zheng Z. Association of PTPN22-C1858T Polymorphism With Susceptibility to Mycobacterium tuberculosis and Mycobacterium leprae Infection: A Meta-Analysis. *Front Immunol*. 2021;12:592841.

CONFERENCE ABSTRACTS

1. Zhang D, Guo Y, Peng Z, Xiao Y, Chang Z, Yu L, Zhao Y, Zhang Q, Ma L, **Li S**, Li CK, Leung KT, Zhao Z, Chen C, Chen Y. FLT3L-based conjugate targets chemoresistant leukemia stem cells via cell cycle re-entry in Acute Myeloid Leukemia. The 67th ASH Annual Meeting and Exposition; November 2025; Orlando, FL, USA. *Blood*; 2025. p. 319.
2. **Li S**, Cheng L, Cui T, Zhang L, Tian X, Zhao ZJ, Chen Y, Zheng Z. High Expression of Mincle in Intermediate Monocytes of Patients with Autoimmune Diseases. ASN Kidney Week 2021; November 4, 2021; Virtual Only. *J Am Soc Nephrol*. 2021. p. 455-6.
3. Liu R, Li X, **Li S**, Ye K, Li S, Yu J, Li L. Effects of Celastrol on expression of DREAM, PSD95 and NR1 in MCAO rats. XXV International Symposium on Morphological Sciences; July 26-30, 2017; Xi'an, China. *Ann Anat*. 2017. p. 216.

SELECTED PRESENTATIONS

- Podium Presentation. OMRF Gene Human Disease Research Program Serial Seminar, October 2026. “Causal Variant Discovery and Functional Characterization in the SLE-risk *STAT4* Locus.”
- Podium Presentation. OMRF Gene Human Disease Research Program Serial Seminar, September 2025. “Causal Variant Discovery and Functional Characterization in the SLE-risk *STAT4* Locus.”

- Poster Presentation and Flash Talk. OUHC GREAT Symposium, 2026. “Causal Variant Discovery and Functional Characterization in the SLE-risk *STAT4* Locus.”
- Poster Presentation. OMRF Retreat, 2026. “Causal Variant Discovery and Functional Characterization in the SLE-risk *STAT4* Locus.”
- Podium Presentation. OMRF Gene Human Disease Research Program Serial Seminar, September 2025. “Causal Variant Discovery and Functional Characterization in the SLE-risk *STAT4* Locus.”
- Poster Presentation. OMRF Retreat, 2025. “Causal Variant Discovery and Characterization in the Systemic Lupus Erythematosus Associated Gene *STAT4*.”
- Poster Presentation. Annual Cancer Research Symposium, OU Health Stephenson Cancer Center, 2023. “CD82 affects the anaerobic glycolysis in cancer stem cell promoting breast cancer proliferation.”
- Abstract Presentation. American Society of Nephrology Kidney Week, 2021. “High Expression of Mincle in Intermediate Monocytes of Patients with Autoimmune Diseases.”

TEACHING EXPERIENCE

- Teaching Assistant, large-scale bioinformatics analysis for medical students rotating in Dr. Zhang's laboratory, University of Oklahoma Health Sciences Center, 01/2023-04/2024.
- Teaching Assistant, clinical diagnosis of lupus nephritis, Department of Nephrology, The Seventh Affiliated Hospital of Sun Yat-sen University, 12/2020.
- Lecture: “Macrophage migration inhibitory factor promotes acute kidney injury,” Annual Forum of Rheumatology in Shenzhen, 12/2019.

HONORS & AWARDS

- Excellent Graduation Thesis, Sun Yat-sen University, 2021.
- Poster Contest Winning Prize, Annual Academic Journal Exhibition Competition, The Seventh Affiliated Hospital of Sun Yat-sen University, 2020.
- Third Prize, Fourth National Basic Medicine Innovation Forum and Experiment Design Competition for College Students, 2016.
- First Prize, Experimental Skills of Morphology Competition, Gannan Medical University, 2015.
- National English Competition for College Students (NECCS), Third Prize, Band C, 2014.

SCHOLARSHIPS & SERVICE

- Scholarships: Sun Yat-sen University (Second-class, 2020; First-class, 2018); Gannan Medical University (2014-2017, multiple first-/second-/third-class scholarships and student honors).
- Committee Member (Treasurer), Oklahoma City Chinese Student and Scholar Association (OKCCSSA), 08/2021-05/2026.

LANGUAGES

English (proficient) | Chinese (native)